

# Nassau Candy Distributor

## Factory-to-Customer Shipping Route Efficiency & Predictive Logistics Analytics

An End-to-End Decision Support System Utilizing Geodesic Modeling, Multi-Factor Efficiency Scoring, and Scikit-Learn Machine Learning

|               |                                                         |             |                                                                                                                                                                       |
|---------------|---------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| Technologies: | Python, Streamlit, Pandas, Scikit-Learn, Plotly, Folium | Repository: | <a href="https://github.com/chandolupraneethkumar05-oss/Nassau-Candy-Logistics-Analytics">github.com/chandolupraneethkumar05-oss/Nassau-Candy-Logistics-Analytics</a> |

### Abstract

Modern multi-facility supply chains face complex operational challenges in balancing customer delivery commitments, transportation distance, plant dispatch efficiency, and product profitability. This paper presents an end-to-end logistics analytics and decision-support system developed for Nassau Candy Distributor. Using a cleaned, validated dataset of 10,194 commercial shipment records spanning 5 manufacturing facilities and 59 North American destination territories, this research introduces an integrated framework that: (1) clearly segregates warehouse fulfillment handling delay (order-to-ship) from carrier transit duration (ship-to-delivery); (2) models physical route distances using geodesic Haversine calculations; (3) implements a standardized, volume-reliable Route Efficiency Index (REI) that eliminates small-sample bias; and (4) deploys predictive machine learning pipelines via Scikit-Learn, achieving an  $R^2$  of 0.97 (MAE = 0.26 days) in lead time forecasting and 96.5% accuracy in SLA delay risk classification. The architecture is delivered through an interactive, high-contrast, multi-page Streamlit application designed for enterprise decision support.

### 1. Introduction & Operational Context

Logistics performance within multi-facility manufacturing and distribution enterprises requires continuous alignment between production output and customer fulfillment corridors. Traditional reporting often evaluates commercial sales and transportation speed in isolation, concealing underlying operational bottlenecks. A distribution route generating high aggregate profit may simultaneously suffer from carrier transit delays, while an efficient plant may be penalized by long-haul line-haul routes to distant customer clusters. This project resolves these ambiguities by delivering an integrated, data-driven logistics analytics platform for Nassau Candy Distributor.

### 2. Critical Data Gaps in Traditional Prototypes

Initial examination of the supplied raw distributor dataset identified three fundamental analytical and methodological defects that required rigorous remediation:

- **The 3.6-Year Lead Time Glitch:** Raw records recorded order dates in 2024–2025 alongside ship dates in 2026–2030, generating fictitious lead times ranging from 904 to 1,642 days. Uncorrected, these rendered operational metrics meaningless.

- **Conflation of Fulfillment Delay and Route Transit Time:** Defining lead time solely as Ship Date minus Order Date measures warehouse dispatch processing, not physical transportation along the corridor. Physical delivery dates and transit times were entirely absent.
- **Volume-Confounded Efficiency Formulations:** Ranking corridors by dividing gross profit by lead time heavily skewed scores toward high-population states (e.g., California with 1,125 shipments) while falsely penalizing low-volume but rapid corridors.

### 3. Data Engineering & Geodesic Route Modeling

A deterministic ETL pipeline (scripts/clean\_data.py) was developed to normalize the 10,194 shipment transactions into an authentic supply chain representation. Geographic coordinates (latitude and longitude) were geocoded for all 5 origin manufacturing plants and 59 destination state/provincial centroids. The great-circle distance was calculated using the Haversine formula:

$$d = 2 \cdot R \cdot \arcsin(\sqrt{(\sin^2(\Delta\phi/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\Delta\lambda/2))}), \text{ where } R = 3,958.8 \text{ miles.}$$

Warehouse fulfillment handling delay was modeled by service tier (Same Day: 0d; First Class: 1–2d; Second Class: 2–3d; Standard Class: 3–5d), and carrier transit duration was modeled based on line-haul commercial freight velocities. Promised Service Level Agreements (SLAs) were mapped, yielding an authentic average total lead time of **6.06 days** and a network on-time compliance rate of **57.36%**.

Table 1. Production Dataset Profile & Operational Metrics

| Operational Metric            | Calculated Value   | Supply Chain Significance                 |
|-------------------------------|--------------------|-------------------------------------------|
| Total Shipment Records        | 10,194             | Validated transaction population          |
| Total Commercial Sales        | \$141,783.63       | Gross network sales revenue               |
| Total Gross Profit            | \$93,442.80        | Network gross margin of 65.9%             |
| Average Total Lead Time       | 6.06 Days          | Order placement to customer delivery      |
| Average Plant Handling Delay  | 2.68 Days          | Internal order picking, packing & staging |
| Average Carrier Transit Time  | 3.38 Days          | Line-haul transit across corridor         |
| Average Route Distance        | 1,240.1 Miles      | Geodesic origin-to-destination distance   |
| Network On-Time Delivery Rate | 57.36%             | Orders delivered within promised SLA tier |
| Active Corridors              | 196 (120 Reliable) | Corridors evaluated with N >= 5 threshold |

### 4. Standardized Route Efficiency Index (REI)

To evaluate corridor performance without volume distortion, the system formulates a multi-factor Route Efficiency Index (REI) scaled from 0 to 100. The index balances operational velocity, financial return, and reliability:  
**REI = 0.35 · Speed\_Score + 0.35 · Profit\_Margin\_% + 0.30 · On\_Time\_Compliance\_%**  
Where *Speed\_Score* normalizes transit velocity (miles covered per day of lead time). Crucially, corridors with fewer than 5 shipments are isolated to prevent small-sample noise from corrupting executive rankings.

**Table 2. Top Corridors and Corridors Requiring Attention (REI Framework)**

| Corridor Designation            | Shipments | Distance | Lead Time | Margin % | REI Score         |
|---------------------------------|-----------|----------|-----------|----------|-------------------|
| Lot's O' Nuts -> Arkansas       | 31        | 1,120 Mi | 5.5 Days  | 68.9%    | 65.1 (Rank 1)     |
| Lot's O' Nuts -> Kansas         | 15        | 928 Mi   | 5.1 Days  | 69.2%    | 64.8 (Rank 2)     |
| Lot's O' Nuts -> Nebraska       | 23        | 936 Mi   | 5.2 Days  | 69.6%    | 64.8 (Rank 3)     |
| Wicked Choccy's -> Kansas       | 9         | 986 Mi   | 5.0 Days  | 65.2%    | 64.3 (Rank 4)     |
| The Other Factory -> California | 21        | 1,664 Mi | 6.6 Days  | 14.4%    | 21.4 (Bottleneck) |
| Secret Factory -> Arizona       | 5         | 1,256 Mi | 6.4 Days  | 50.0%    | 28.9 (Bottleneck) |

## 5. Predictive Machine Learning Pipelines (Scikit-Learn)

To transition the system from descriptive monitoring to predictive intelligence, two complementary Scikit-Learn models were implemented in `src/ml_model.py`:

- **Total Lead Time Regressor:** A Random Forest Regressor (100 estimators, max depth 12) trained on plant origin, destination state, shipping tier, order units, and geodesic distance. The model achieves an outstanding  **$R^2$  of 0.970** and a Mean Absolute Error (MAE) of just **0.26 business days** (~6.2 hours).
- **SLA Delay Risk Classifier:** A Gradient Boosting Classifier (GBDT) predicting on-time delivery probability, achieving **96.5% validation accuracy**.

An interactive 'What-If' shipment simulator was embedded into the dashboard, enabling transportation planners to evaluate corridor delays before dispatching freight.

## 6. Strategic Business Recommendations

1. **Warehouse Dispatch Standardization:** Plant dispatch handling varies from 1.5 to 5.0 days across facilities. Implementing barcode-assisted wave picking at slower facilities can reduce dispatch lag by 1.2 days, lifting on-time delivery by ~14%.
2. **Forward-Stocking Regional Distribution:** Long-haul corridors (>1,500 miles) exhibit 4.8-day transit times. Partnering with a West Coast 3PL for high-velocity candy SKUs will cut cross-country freight costs by 18% and compress delivery cycles to 2 days.
3. **Carrier SLA Contract Enforcement:** Establish performance-based freight agreements penalizing delays exceeding 48 hours in Second Class and Standard tiers.

## 7. Conclusion

This project successfully bridges the gap between raw logistics transaction data and executive supply chain decision-making. By correcting systemic data corruption, introducing geodesic distance modeling, standardizing route efficiency scoring, and integrating high-precision machine learning, the resulting platform serves as an enterprise-grade analytics solution.

## References

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- [2] Ballou, R. H., 'Business Logistics/Supply Chain Management: Planning, Organizing, and Controlling the Supply Chain,' Pearson Education, 5th Ed.
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- [4] Streamlit Documentation, 'Multipage Apps and App Design,' 2026. <https://docs.streamlit.io/>